Yash Verma

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Summary

Senior Unreal Engine C++ Programmer with 11+ years of experience developing and shipping multiplayer games, including Squad 44, Path of Titans, and Aquanox Deep Descent. Specializing in server-authoritative networking, gameplay systems, vehicle physics, engine-level development, and performance optimization. Comfortable working independently on complex gameplay systems, from initial design through implementation and release, as well as collaborating with distributed teams remotely.

Skills & Abilities

Engine: Unreal Engine 4/5, Unreal Gameplay Framework, engine source modifications, plugins/modules
Programming: C++, Blueprints, gameplay architecture, Asynchronous programming, multithreading
Multiplayer: Client-side prediction, Replication, Server Authority validation, Online Backend Integrations
Gameplay: Combat, movement, vehicles, UI/UMG, physics, animation, AI, persistence/save systems
Engineering: Profiling, debugging, code reviews, optimization, documentation
Tools: Perforce, Git, SVN, Visual Studio, Unreal Insights, Superluminal
Additional: SDK integration, online services, external hardware, embedded systems

Achievements

- Founded and led Heroes of the West, securing official support from Tripwire Interactive and achieving 40,000 downloads in its first month on Steam.
- Led the programming team on Squad 44 (formerly Post Scriptum), contributing to the delivery of multiple game modes and major gameplay systems.
- Designed and implemented advanced multiplayer combat architecture featuring rollback hit validation, projectile ballistics, and modular weapon systems.
- Developed a custom dual-transmission tank drivetrain using Chaos physics, overcoming limitations in Unreal Engine's existing vehicle framework.

Experience

SENIOR GAMEPLAY PROGRAMMER | ALDERON GAMES (REMOTE) | JANUARY 2023 – AUGUST 2026

Unannounced project:

- Engineered end-to-end networked combat systems, including projectile ballistics, rollback hit validation, animation integration, and modular weapon and attachment customization.
- Developed systemic gameplay features for large-scale worlds, including dynamic content spawning and persistent item save/load systems for partitioned environments, targeted for 100+ concurrent players.
- Architected and implemented a replicated character attribute system supporting progression, efficient stat modifier evaluation, multiple damage types, and extensible damage processing.
- Built a replicated, modular status effect system supporting attribute modifiers, timed/event-driven expiration.
- Designed and implemented character customization systems and supporting gameplay and UI functionality.
- Developed vehicle component damage systems and associated gameplay mechanics.

Path of Titans:

- Implemented server-authoritative validation to improve multiplayer integrity and reduce exploit risks.
- Developed and maintained UI features and gameplay systems using Unreal Engine's Gameplay Ability System (GAS).
- Profiled and optimized gameplay systems, reducing execution time for targeted operations from approximately 2 ms to a few hundred microseconds.
- Diagnosed and resolved gameplay, UI, networking, and system-level defects.

TECHNICAL LEAD | PERISCOPE GAMES (REMOTE) | JUNE 2018 – DECEMBER 2022

Squad 44 (Formerly Post Scriptum)

- Led the programming team, coordinating development efforts, assigning responsibilities, and overseeing technical implementation across gameplay systems.
- Developed and shipped multiple game modes, including Invasion, Sandbox, and Zombie.
- Built an armored vehicle combat system encompassing projectile ballistics, armor penetration, damage modeling, driving physics, crew seating, and vehicle interactions.
- Implemented gameplay features including character and vehicle movement, melee combat, weapons, equipment, ragdolls, gore and dismemberment systems, map voting, and camera systems.
- Completed and integrated the game's modding framework, including UI integration, Steam backend connectivity, bug fixing, and improvements to the mod development pipeline.
- Designed and developed rule-based code generation tools to automate repetitive programming tasks and streamline development workflows.
- Performed engine upgrades, performance optimization, debugging, and general systems maintenance.

Unannounced Project

- Developed custom tracked vehicle driving physics by extending Unreal Engine's Chaos Vehicles framework.
- Implemented vehicle interior seating and interaction systems, including console controller integration.

PROGRAMMER | DIGITAL ARROW | MARCH 2017 – APRIL 2018

Worked as a gameplay programmer on *Aquanox Deep Descent*, published by THQ Nordic.

- Developed and refined player character movement systems and physics interactions.
- Contributed to NPC AI behaviour, damage systems, and animation integration.
- Diagnosed and resolved gameplay bugs across various systems.

TEAM LEADER | HEROES OF THE WEST, KERNOW INTERACTIVE | DEC 2014 – JAN 2017

Founded and led development of Heroes of the West, a total conversion mod for Red Orchestra 2, officially supported and published by Tripwire Interactive on Steam.

- Recruited and led the development team, overseeing engineering, production, and technical direction.
- Led gameplay programming, implementing new features and systems.
- Developed internal tools for code generation and asset processing to streamline development workflows.
- Managed builds, source control, Steam distribution, and post-release updates.

Education

BACHELOR OF ENGINEERING | COMPUTER SCIENCE AND ENGINEERING | B.I.T DURG

- Graduated from 2014 batch.